Alberto Fachechi

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Activity and research interests

Contents: My research activity is placed in the wide discipline of statistical mechanics methods for complex systems analysis, in particular concerning spin-glass theory and its application to modern Artificial Intelligence and Machine Learning. Within this framework, I also study data-driven associative memories, i.e. networks that build their memories from examples, addressing how they capture the low-dimensional structure of data and how this affects retrieval, generalization and memorization.

Methods: The core research interest consists in the development of rigorous mathematical techniques, involving in particular Guerra's interpolating schemes, non-linear PDE theory, probability and statistics and random matrix theory.

Other interests: My research interests also cover the application of statistical inference tools to real-world (in particular, biological) problems.

Positions

Apr 2026 – today
- **Tenure-Track Researcher**
  Department of Mathematics "Guido Castelnuovo", Sapienza Università di Roma, Rome, Italy
  SSD: MAT/07 (now MATH-04/A) - Mathematical Physics

Apr 2023 – Mar 2026
- **Researcher** (RTD-A in national denomination)
  Department of Mathematics "Guido Castelnuovo", Sapienza Università di Roma, Rome, Italy
  SSD: MAT/07 (now MATH-04/A) - Mathematical Physics

Sep 2021 – Apr 2023
- **Post-doc researcher**
  Department of Mathematics "Guido Castelnuovo", Sapienza Università di Roma, Rome, Italy
  SSD: MAT/07 (now MATH-04/A) - Mathematical Physics

Mar 2020 – Feb 2021
- **Post-doc researcher**
  Department of Mathematics and Physics "Ennio De Giorgi", Università del Salento, Lecce, Italy
  SSD: MAT/07 (now MATH-04/A) - Mathematical Physics

Feb 2019 – Jan 2020
- **Research fellow**
  Department of Mathematics and Physics "Ennio De Giorgi", Università del Salento, Lecce, Italy
  SSD: MAT/07 (now MATH-04/A) - Mathematical Physics

Education

Jan 2016 – Jan 2019
- **PhD**
  Physics and Nanosciences, Department of Mathematics and Physics "Ennio De Giorgi", Università del Salento, Lecce, Italy
  Thesis: *Statistical mechanics for Artificial Intelligence: Learning, Retrieving, Unlearning and Sleeping*

Oct 2013 – Oct 2015
- **MSc**
  Theoretical and Fundamental Interactions Physics, Department of Mathematics and Physics "Ennio De Giorgi", Università del Salento, Lecce, Italy
  Thesis: *Higher-spin Lifshitz theories and integrable systems*

Education (continued)

Sep 2009 – Oct 2013
- **BSc**
  Physics, Department of Mathematics and Physics "Ennio De Giorgi", Università del Salento, Lecce, Italy
  Thesis: *Integrability in AdS/CFT*

Scientific communities

2018 – today
- **National Group of Mathematical Physics (GNFM-INdAM)**
  Section of Lecce (2018-2021) and Rome (2021-Today)
  Member
  Subsection: Mechanics of discrete systems

2016 – 2017
- **GAuge Theory as Integrable System (GATIS)**
  INFN Section of Bologna
  Member as PhD Student
  Coordinators: Matteo Beccaria, Davide Fioravanti, Marco Rossi

2015 – 2021
- **National Institute of Nuclear Physics (INFN)**
  Section of Theoretical Physics (CSN4), Lecce
  Member

Teaching

Oct 2026 – Jul 2027
- **Lecturer**
  Calculus and Probability, Department of Biology, Sapienza Università di Roma, Rome (IT)
- **Teacher Assistant**
  Probability and Statistics in High Dimension, Department of Mathematics, M.Sc. in Applied Mathematics, Sapienza Università di Roma, Rome (IT)

Oct 2025 – Jul 2026
- **Lecturer**
  Statistics, Department of Mathematics, Sapienza Università di Roma, Rome (IT)
- **Teacher Assistant**
  Probability and Statistics in High Dimension, Department of Mathematics, M.Sc. in Applied Mathematics, Sapienza Università di Roma, Rome (IT)

Oct 2024 – Jul 2025
- **Lecturer**
  Mathematics, Department of Biology and Biotechnologies, B.Sc. in Biotechnologies, Sapienza Università di Roma, Rome (IT)

Mar – May 2024
- **Lecturer**
  Statistical learning and neural networks: a gentle introduction to modern Artificial Intelligence, Department of Mathematics, Excellence program for the B.Sc. in Mathematics, Sapienza Università di Roma, Rome (IT)

Nov – Dec 2024
- **Teacher assistant**
  Elements of Probability and Statistics for Data Science, Department of Mathematics, M.Sc. in Applied Mathematics, Sapienza Università di Roma, Rome (IT)

Oct 2023 – Jul 2024
- **Co-lecturer**
  Calculus and Biostatistics, Department of Biology and Biotechnologies, B.Sc. in Biology, Sapienza Università di Roma, Rome (IT)

Nov 2021 – Nov 2022
- **Co-lecturer**
  Elements of Mathematics, Department of Architecture, B.Sc. in Architecture, Sapienza Università di Roma, Rome (IT)

Teaching (continued)

Nov 2020 – Mar 2021
- **Teaching Assistant (exercise sessions)**
  Mathematics for Economy and Finance, Department of Economy, B.Sc. in Economy and Finance, Unisalento, Lecce (IT)

Feb – Jul 2020
- **Teaching Assistant (exercise sessions)**
  Mathematics for Economy and Finance, Department of Economy, B.Sc. in Economy and Finance, Unisalento, Lecce (IT)

Supervising

- Supervisor or co-supervisor of 4 B.Sc., 6 M.Sc. and 3 Ph.D. theses (2 ongoing).

Funded projects

2023
- **Rigorous statistical mechanics for Artificial Intelligence and Machine Learning**
  Role: PI
  Project funded by the National Group of Mathematical Physics (GNFM-INdAM) in the context of "Progetto Giovani 2023".

2022
- **Large scale multicomponent random systems**
  Role: partecipant
  Ref: Prof. Vittoria Silvestri
  Project funded by Sapienza Università di Roma as "Progetto di Ricerca Medio 2022"
- **Replica-symmetry breaking in modern Artificial Intelligence**
  Role: PI
  Project funded by Sapienza Università di Roma as "Progetto di Avvio alla Ricerca 2022"

2018
- **Rigorous approach to Deep Learning**
  Role: partecipant
  Ref: Prof. Elena Agliari
  Project funded by the National Group of Mathematical Physics (GNFM-INdAM) in the context of "Progetto Giovani 2018".

Other activities

Organization

Jul 2025 – Jan 2026
- **Member of the local Organizing Committee of the 4th Workshop of UMI Group "Mathematics for Artificial Intelligence and Machine Learning"**
  Department of Mathematics G. Castelnuovo, Sapienza Università di Roma
  Fourth edition of the workshop of the UMI Group "Mathematics for Artificial Intelligence and Machine Learning", following the previous ones held in Milan, Turin, and Bari.

Nov 2024 – Sep 2025
- **Co-organizer of the Summer School "Mathematical methods for high-dimensional data"**
  Department of Mathematics G. Castelnuovo, Sapienza Università di Roma
  The school will open the thematic period on Data Science of the "Excellence program" for the Department of Mathematics: see the website.

2024 – today
- **Responsible for communication**
  Communication roles for the FAIR foundation, Spoke 5 (WP5.5) for the organization of scientific and dissemination events.

Other activities (continued)

2024 — **Organizer of scientific meetings**
I covered organizing roles for the FAIR Spoke 5 scientific meeting, consisting in discussions and crossfertilization between working packages for the FAIR (Future Artificial Intelligence Research).

Other roles

2022 — **Project reviewer**
I have also been a project reviewer for the ISF

2017 – today — **Referee for high-impact journals**
Among others: *Physical Reviews X; IEEE Transaction on Neural Networks and Learning Systems; Nature Scientific Reports; Journal of Mathematical Physics; Journal of Physics A: Mathematical and Theoretical; Neural Computation; Journal of Physics: Complexity; Physica A: Statistical Mechanics and its Applications; Journal of Statistical Mechanics: Theory and Experiment; Machine Learning: Science and Technology; Physical Review E*

Visiting periods

16 – 23 Jan 2023 — **Alan Turing Institute, London (UK)**
During the period, I attended the workshop "Physics-informed Machine-Learning", in which I took part to stimulating discussions about statistical mechanics of spin glasses and their application to Artificial Intelligence. I also gave a talk at the workshop about Dreaming Neural Networks and the possibility to extended the theory to Boltzmann Machines.

1 – 5 Oct 2018 — **Department of Mathematics, King's College London, London (UK)**
Ref: Prof. Alessia Annibale
During the period, we discussed several topics in statistical mechanics of spin glasses, especially concerning the application of Machine Retrieval models to the theoretical modeling of biological processes. In this period, I also gave an invited talk. Finally, I also had the chance to discuss with brilliant researchers and professors, in particular with Prof. Reimer Kuhn.

2018 (often) — **Department of Mathematics "Guido Castelnuovo", Sapienza Università di Roma, Rome (IT)**
Ref: Prof. Elena Agliari
During the period, we investigated unlearning theory for Machine Retrieval models. The discussions were crucial for the ideation and the theoretical investigation about Dreaming Neural Networks.

2016 – 2017 (often) — **Department of Physics "Augusto Righi", Alma Mater Studiorum, Bologna (IT)**
Ref: Prof. Davide Fioravanti
During the period, we investigated the relation between supersymmetric gauge theory on deformed background with quantization of classical integrable systems.

Summary of scientific production

- Total number of published papers: 33
- Total number of papers: 36
- Total number of citations[1]: 733
- h-index[1]: 12
- $i10$-index[1]: 14

[1] Source: GoogleScholar.

Publications

Journal articles

[1] E. Agliari, S. Akter, A. Alessandrelli, and **A. Fachechi**, "Pattern disentanglement and reconstruction by modular Hebbian networks," *Bollettino dell'Unione Matematica Italiana*, 2026. DOI: 10.1007/s40574-026-00538-2.

[2] E. Agliari, P. Duarte Mourao, and **A. Fachechi**, "The beneficial role of noises for disentanglement tasks in modular Hebbian networks," *Journal of Physics A: Mathematical and Theoretical*, vol. 682, p. 131 134, 2026. URL: https://www.sciencedirect.com/science/article/pii/S0378437125007861.

[3] E. Agliari, **A. Fachechi**, and D. Luongo, "The importance of being empty: a spectral approach to Hopfield neural networks with diluted examples," *Neurocomputing*, vol. 694, p. 133 789, 2026. URL: https://www.sciencedirect.com/science/article/pii/S0925231226011860.

[4] **A. Fachechi**, E. Agliari, M. Aquaro, A. Coolen, and M. Mulder, "Fundamental operating regimes, hyper-parameter fine-tuning and glassiness: towards an interpretable replica-theory for trained restricted Boltzmann machines," *Journal of Physics A: Mathematical and Theoretical*, vol. 58, p. 065 004, 2025. DOI: 10.1088/1751-8121/ada986.

[5] E. Agliari, F. Alemanno, M. Aquaro, and **A. Fachechi**, "Regularization, early-stopping and dreaming: a Hopfield-like setup to address generalization and overfitting," *Neural Networks*, vol. 177, p. 106 389, 2024. URL: https://www.sciencedirect.com/science/article/pii/S0893608024003137.

[6] E. Agliari, **A. Fachechi**, and D. Luongo, "A spectral approach to Hebbian-like neural networks," *Applied Mathematics and Computation*, vol. 474, p. 128 689, 2024. URL: https://www.sciencedirect.com/science/article/pii/S0096300324001619.

[7] J. Steinberg, U. Adomaityte, **A. Fachechi**, P. Mergny, D. Barbier, and R. Monasson, "Replica method for computational problems with randomness: principles and illustrations," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2024, p. 104 002, 2024, Lectures by R Monasson given at the summer school 'Statistical Physics & Machine Learning', which was organised by F Krzakala and L Zdeborova and took place in Les Houches School of Physics in France from 4 to 29 July 2022. DOI: 10.1088/1742-5468/ad292d.

[8] E. Agliari, M. Aquaro, A. Barra, **A. Fachechi**, and C. Marullo, "From Pavlov conditioning to Hebb learning," *Neural Computation*, vol. 35, no. 5, pp. 930–957, 2023. URL: https://direct.mit.edu/neco/article-abstract/35/5/930/115251/From-Pavlov-Conditioning-to-Hebb-Learning.

[9] F. Alemanno, M. Cavo, D. Delle Cave, **A. Fachechi**, R. Rizzo, E. D'Amone, G. Gigli, E. Lonardo, A. Barra, and L. L. Del Mercato, "Quantifying heterogeneity to drug response in cancer–stroma kinetics," *Proceedings of the National Academy of Sciences*, vol. 120, no. 11, e2122352120, 2023. DOI: 10.1073/pnas.2122352120.

[10] E. Agliari, **A. Fachechi**, and C. Marullo, "Nonlinear PDEs approach to statistical mechanics of dense associative memories," *Journal of Mathematical Physics*, vol. 63, no. 10, 2022. URL: https://pubs.aip.org/aip/jmp/article/63/10/103304/2843264.

[11] **A. Fachechi**, A. Barra, E. Agliari, and F. Alemanno, "Outperforming RBM feature-extraction capabilities by "dreaming" mechanism," *IEEE transactions on neural networks and learning systems*, vol. 35, no. 1, pp. 1172–1181, 2022. URL: https://ieeexplore.ieee.org/document/9801668.

[12] E. Agliari, L. Albanese, F. Alemanno, and **A. Fachechi**, "A transport equation approach for deep neural networks with quenched random weights," *Journal of Physics A: Mathematical and Theoretical*, vol. 54, no. 50, p. 505 004, 2021. DOI: 10.1088/1751-8121/ac38ec.

[13] **A. Fachechi**, "PDE/Statistical Mechanics Duality: Relation Between Guerra's Interpolated p-Spin Ferromagnets and the Burgers Hierarchy," *Journal of Statistical Physics*, vol. 183, no. 1, p. 12, 2021. DOI: 10.1007/s10955-021-02747-9.

[14] E. Agliari, F. Alemanno, A. Barra, O. A. Barra, **A. Fachechi**, L. F. Vento, and L. Moretti, "Analysis of temporal correlation in heart rate variability through maximum entropy principle in a minimal pairwise glassy model," *Scientific Reports*, vol. 10, no. 1, p. 15 353, 2020. URL: https://www.nature.com/articles/s41598-020-72183-4.

[15] E. Agliari, F. Alemanno, A. Barra, M. Centonze, and **A. Fachechi**, "Neural networks with a redundant representation: Detecting the undetectable," *Physical review letters*, vol. 124, no. 2, p. 028 301, 2020. DOI: 10.1103/PhysRevLett.124.028301.

[16] E. Agliari, F. Alemanno, A. Barra, and **A. Fachechi**, "Generalized Guerra's interpolation schemes for dense associative neural networks," *Neural Networks*, vol. 128, pp. 254–267, 2020. URL: https://www.sciencedirect.com/science/article/pii/S0893608020301714.

[17] E. Agliari, A. Barra, O. A. Barra, **A. Fachechi**, L. Franceschi Vento, and L. Moretti, "Detecting cardiac pathologies via machine learning on heart-rate variability time series and related markers," *Scientific Reports*, vol. 10, no. 1, p. 8845, 2020. URL: https://www.nature.com/articles/s41598-020-64083-4.

[18] E. Agliari, **A. Fachechi**, and C. Marullo, "The relativistic Hopfield model with correlated patterns," *Journal of Mathematical Physics*, vol. 61, no. 12, 2020. URL: https://pubs.aip.org/aip/jmp/article/61/12/123301/318057.

[19] F. Alemanno, M. Centonze, and **A. Fachechi**, "Interpolating between Boolean and extremely high noisy patterns through minimal dense associative memories," *Journal of Physics A: Mathematical and Theoretical*, vol. 53, no. 7, p. 074 001, 2020. DOI: 10.1088/1751-8121/ab6943.

[20] E. Agliari, F. Alemanno, A. Barra, and **A. Fachechi**, "Dreaming neural networks: rigorous results," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2019, no. 8, p. 083 503, 2019. DOI: 10.1088/1742-5468/ab371d.

[21] E. Agliari, F. Alemanno, A. Barra, and **A. Fachechi**, "On the Marchenko–Pastur law in analog bipartite spin-glasses," *Journal of Physics A: Mathematical and Theoretical*, vol. 52, no. 25, p. 254 002, 2019. DOI: 10.1088/1751-8121/ab1934.

[22] **A. Fachechi**, E. Agliari, and A. Barra, "Dreaming neural networks: forgetting spurious memories and reinforcing pure ones," *Neural Networks*, vol. 112, pp. 24–40, 2019. URL: https://www.sciencedirect.com/science/article/pii/S0893608019300176.

[23] E. Alfinito, A. Barra, M. Beccaria, **A. Fachechi**, and G. Macorini, "An evolutionary game model for behavioral gambit of loyalists: Global awareness and risk-aversion," *Europhysics Letters*, vol. 121, no. 3, p. 38 001, 2018. DOI: 10.1209/0295-5075/121/38001.

[24] A. Barra, M. Beccaria, and **A. Fachechi**, "A new mechanical approach to handle generalized Hopfield neural networks," *Neural Networks*, vol. 106, pp. 205–222, 2018. URL: https://www.sciencedirect.com/science/article/pii/S0893608018302090.

[25] **A. Fachechi**, G. Macorini, and M. Beccaria, "Chiral Trace Relations in $\mathcal{N} = 2^*$ Supersymmetric Gauge Theories," *Theoretical and Mathematical Physics*, vol. 196, no. 3, pp. 1282–1293, 2018. DOI: 10.1134/s0040577918090039.

[26] **A. Fachechi**, G. Macorini, and M. Beccaria, "Chiral trace relations in Ω-deformed $\mathcal{N} = 2$ theories," *Journal of Physics: Conference Series*, vol. 965, no. 1, p. 012 013, 2018. DOI: 10.1088/1742-6596/965/1/012013.

[27] E. Alfinito, M. Beccaria, **A. Fachechi**, and G. Macorini, "Reactive immunization on complex networks," *Europhysics Letters*, vol. 117, no. 1, p. 18 002, 2017. DOI: 10.1209/0295-5075/117/18002.

[28] M. Beccaria, **A. Fachechi**, and G. Macorini, "Chiral trace relations in Ω-deformed $\mathcal{N} = 2$ theories," *Journal of High Energy Physics*, vol. 2017, no. 5, pp. 1–37, 2017. DOI: 10.1007/JHEP05(2017)023.

[29] M. Beccaria, **A. Fachechi**, and G. Macorini, "On the cusp anomalous dimension in the ladder limit of $N = 4$ SYM," *Journal of High Energy Physics*, vol. 2016, no. 6, pp. 1–21, 2016. DOI: 10.1007/JHEP06(2016)009.

[30] M. Beccaria, **A. Fachechi**, and G. Macorini, "Virasoro vacuum block at next-to-leading order in the heavy-light limit," *Journal of High Energy Physics*, vol. 2016, no. 2, pp. 1–22, 2016. DOI: 10.1007/JHEP02(2016)072.

[31] M. Beccaria, **A. Fachechi**, G. Macorini, and L. Martina, "Exact partition functions for Ω-deformed $\mathcal{N} = 2$ theories with $N_f = 4$ flavours," *Journal of High Energy Physics*, vol. 2016, no. 12, pp. 1–41, 2016. DOI: 10.1007/JHEP12(2016)029.

Conference proceedings

[1] E. Alfinito, M. Beccaria, **A. Fachechi**, and G. Macorini, "Probing complexity with epidemics: a new reactive immunization strategy," in *COMPLEXIS 2017, 2nd International Conference on Complexity, Future Information Systems and Risk*, vol. 1, 2017, pp. 116–123. URL: https://www.scitepress.org/ProceedingsDetails.aspx?ID=5x88vx+dW4Q=&t=1.

Preprints

[1] E. Agliari, A. Alessandrelli, A. Barra, and **A. Fachechi**, *Dense auto-hetero associative memories applied to noisy communication channels*, arXiv preprint, 2026. arXiv: 2609.32605.

[2] E. Agliari, P. Duarte Mourão, **A. Fachechi**, and P. Vivo, *Spectral criteria for generalization in unsupervised Hebbian nets*, arXiv preprint, 2026. arXiv: 2606.04651.

[3] E. Agliari, A. Alessandrelli, P. Duarte Mourao, and **A. Fachechi**, *Multi-channel pattern reconstruction through L-directional associative memories*, arXiv preprint, 2025. arXiv: 2503.06274.

[4] E. Agliari, A. Barra, P. Bianco, **A. Fachechi**, and D. Pallara, *The thermodynamic limit in mean field neural networks*, arXiv preprint, 2024. arXiv: 2409.10145.

Selected talks, invited talks and courses

Nov 2026 — **Order vs. disorder in neural networks: a journey through memories and forgetting**
Short course on associative memories at ArpiLYSM3, Arpino (IT)

Oct 2026 — **A journey through memories: From information retrieval to emergence of concepts in associative neural networks throught the lens of equilibrium statistical mechanics**
Invited talk at the national council of GNFM, Montecatini Terme (IT)

Jun 2026 — **Dense L-directional Hopfield networks: merging modularity and higher-order interactions for pattern disentanglement**
Invited talk at the workshop "Mathematical Foundations of AI", Chalmers Institute of Technology (SWE)

Jan 2026 — **The thermodynamic limit in mean-field neural networks**
Invited talk at the conference "Guerra Legacy: The role of Data Science" in memory of Prof. Francesco Guerra, Sapienza Università di Roma (IT)

Sep 2025 — **Associative memory in neural networks: What is memory?**
Invited talk at the event "Is AI intelligence?", Superior School of Advanced Studies, Sapienza Università di Roma (IT)

Selected talks, invited talks and courses (continued)

Apr 2025
- **Statistical mechanics of equilibrium of learning Boltzmann Machines: fundamental operating regimes, hyper-parameter fine-tuning and glassiness**
  Invited talk at Chimera Journal Club, Physics Department, Sapienza Università di Roma (IT)

Jan 2025
- **A random matrix approach to Hopfield-like neural networks: addressing generalization and overfitting**
  3rd Workshop of UMI Group Mathematics for Artificial Intelligence and Machine Learning, Politecnico di Bari (IT)

Nov 2024
- **Modeling information processing systems: a survey of statistical-mechanical results for bio-inspired neural networks**
  Online invited talk for Dipoppa Lab, Neuroscience group at UCLA (hosted by Matteo Mariani, PhD)

Sep 2024
- **Fundamental operating regimes, hyper-parameter fine-tuning and glassiness: towards an interpretable replica-theory for trained restricted Boltzmann machines**
  ROC(K)IN' AI, Roccella Jonica (IT)

Jan 2024
- **A Hopfield-like picture to address generalization and overfitting**
  Mathematics for Artificial Intelligence and Machine Learning, University Bocconi of Milano, Milano (IT)

May 2023
- **Retrieving, unlearning and learning: there and back again**
  Invited talk at the Department of Physics, Sapienza Università di Roma, Department of Physics, Sapienza Università di Roma, Rome (IT)
- **Machine Learning from a spin-glass theory perspective**
  Invited talk at the workshop Stochastic Models for Complex Systems (SMOCS2023), Unisalento, Unisalento, Lecce (IT)

Jan 2023
- **Dreaming Neural Networks: pushing retrieval/learning capabilities to the limit by "sleeping" mechanism**
  Invited talk at the workshop Physics-informed machine learning workshop, Alan Turing Institute, Alan Turing Institute, London (UK)

Nov 2022
- **Dreaming Neural Networks: pushing retrieval/learning capabilities to the limit by "sleeping" mechanism**
  Matematica per l'Intelligenza Artificiale e il Machine Learning - Giovani ricercatori, Politecnico di Torino, Turin (UK)

Jul 2022
- **Dreaming Neural Networks: pushing retrieval/learning capabilities to the limit by "sleeping" mechanism**
  Invited talk at Unconventional Computation Workshop, organized at the Swansea University (WL), Online talk

Jun 2022
- **PDE/Statistical Mechanics relation: from p-spin ferromagnets to dense associative memories**
  Invited talk at Department of Mathematics, Sapienza Università di Roma, Department of Mathematics, Sapienza Università di Roma, Rome (IT)

Nov 2021
- **Dreaming Neural Networks: dal richiamo automatico all'apprendimento**
  Invited talk for DFNM (Divulgazioni Notturne di Fisica Matematica), organized by GNFM, Online talk

Sep 2021
- **Dreaming Neural Networks: from retrieving to learning,**
  First Young Applied Mathematicians Conference (YAMC2021), Leuca, Lecce (IT)

Oct 2018
- **Sleeping in Hopfield neural networks: some recent results**
  Invited talk at Department of Mathematics, King's College London, London (UK)

Selected talks, invited talks and courses (continued)

Jun 2017

- **Chiral trace relations in Ω-deformed $\mathcal{N} = 2$ theories**
  Physics and Mathematics of Nonlinear Phenomena - "50 years of IST" (PMNP2017), Gallipoli, Lecce (IT)
- **Chiral trace relations in Ω-deformed $\mathcal{N} = 2$ theories**
  XXVth International Conference on Integrable Systems and Quantum symmetries, Czech Technical University Prague, Prague (CZ)